

The schemes, applied throughout: a binary answer with a missing leading zero loses the mark for that value (the register is eight wheels; N21/13 gives one mark of two for the leading zeros alone); working is the worths added; "because computers only understand 1 and 0" is the wrong answer to why binary — the switch is the reason (M24/12).

1. March 2021 P12 Q1(a)(b)(c) — the hockey club's register. [2 + 1 + 1] (Published scheme.)

- (a) [2] one mark per register — $46 = 00101110$ ($32 + 8 + 4 + 2$) · $171 = 10101011$ ($128 + 32 + 8 + 2 + 1$).
 (b) [1] — 255.
 (c) [1] — 11 bits (ten wheels reach 1023; the eleventh first turns at 1024, and $2000 < 2048$). Accept the working " $2^{11} = 2048$ " or "1024 is not enough, 2048 is".

2. Ours — the four-wheel counter. [2 + 1]

- (a) [2] — 0110 · 0111 · 1000 · 1001: two marks for all four in order, one for the first two (0110, 0111). The commonest slip is 0111 → 1111 (no carry); the count says 1 has no next face, so write 0 and carry — three times.
 (b) [1] — the press from 0111 to 1000 turns the fourth wheel for the first time; it is worth 8. "The fourth wheel" alone: 0 — the worth is asked.

3. Ours — the cards. [1 + 2]

- (a) [1] — cards ①, ③, ⑥ are worth 1, 4 and 32: 37.
 (b) [2] — $45 = 32 + 8 + 4 + 1$, so cards ⑥, ④, ③, ① (any order) [1]; 00101101 [1] — six wheels show 101101, and the register's two unturned wheels in front are written. 101101 without the two zeros: the second mark is lost.

4. Ours — read the wheels. [2]

- (a) 01010101 = $64 + 16 + 4 + 1 = 85$ · (b) 11101011 = $128 + 64 + 32 + 8 + 2 + 1 = 235$. One mark each; the worths written earn the mark even with a slip in the sum.

5. Ours — write the wheels. [4]

- (a) $53 = 32 + 16 + 4 + 1 \rightarrow 00110101$ · (b) $170 = 128 + 32 + 8 + 2 \rightarrow 10101010$. One mark for the working (the worths, or the $\div 2$ ladder), one for the eight-wheel answer, each.

6. Ours — sixteen wheels. [2] (N21/13 Q2's shape — the deck's real thing; different numbers here.)

- $300 = 256 + 32 + 8 + 4 \rightarrow 100101100$ [1] → all sixteen wheels: 0000000100101100 [1]. The second mark is the leading zeros, exactly as the real question marks it.

7. Ours — why binary. [2] (M24/12 Q1(a)'s answer, in the student's words.)

- a switch has **two states, on or off** — a wheel with two faces [1]; so each switch can hold **one binary digit, 1 or 0**, and data is represented as patterns of those digits / a base with ten faces would need a switch with ten reliable positions, which is not what the hardware gives [1]. "Computers only understand 0 and 1": 0 — it restates the question.

🕒 **The quiet one, unhinted.** Question 4(a) is 01010101 = 85 and question 5(b) is 170 = 10101010 — **exact flips of each other**, every 1 where the other has a 0 — and $85 + 170 = 255$, **question 1(b)**. Question 2 is the deck's two-faced odometer, 3 its cards, 6 its sixteen-wheels slide.

🔢 **The dig, computed:** card ⑦ is every number from **64 to 127** — 64 numbers, first number **64** (the seventh wheel first turns at 64); seven cards read minds up to **127** ($1 + 2 + 4 + 8 + 16 + 32 + 64$). Card ⑧ would start at **128** and reach 255 — the byte.